

Vueports Solutions — Database Schema Reference

Table Count: 26 tables across 9 domains

1. Core & Authentication (4 tables)

Table	Purpose	Key Fields
clients	Client portal accounts	id, email, password, company, status
admins	Admin dashboard users	id, username, role, 2fa_enabled
admin_sessions	JWT session tracking	token, expires_at, ip_address
password_resets	Token-based password reset	token, user_type, expires_at

2. Site Configuration (1 table)

Table	Purpose
settings	Key-value app configuration with groups

3. Content & Marketing (3 tables)

Table	Purpose	SEO Fields
services	Service catalog pages	slug, meta_title, meta_description
blog_posts	CMS blog articles	slug, published_at, view_count
portfolio_items	Project showcase	slug, technologies, testimonial

4. Projects & Collaboration (4 tables)

Table	Purpose	Status Values
projects	Client project records	planning, in_progress, review, completed, on_hold, cancelled
project_milestones	Project phase tracking	pending, in_progress, completed, blocked
project_files	File attachments	filename, stored_name, mime_type
project_comments	Discussion thread	client_id / admin_id, is_internal

5. Financial (4 tables)

Table	Purpose	Compliance
invoices	Client billing	invoice_number, tax_amount, total_amount (generated)
invoice_items	Line items per invoice	quantity, unit_price, total (generated)
payments	Payment records	No card data —PCI-DSS via PayFast
subscriptions	Recurring billing	billing_cycle, next_billing_date

6. Leads & CRM (4 tables)

Table	Purpose	Source Tracking
consultations	Lead capture from forms	source, status pipeline, ip_hash
bookings	Meeting scheduler	booking_date, meeting_link, calendar_event_id
messages	Contact form submissions	status, replied_at, replied_by
calculator_leads	Cost estimator leads	estimated_total, features JSON

7. Email System (3 tables)

Table	Purpose
email_queue	Async send queue with retry logic
email_logs	Delivery tracking (opened, clicked, bounced)
email_templates	Reusable HTML/text templates with variables

8. Analytics & Monitoring (2 tables)

Table	Purpose	Privacy
page_views	Traffic analytics	ip_hash (not raw IP), no PII
events	Custom event tracking	user_type, event_data JSON

9. Audit & Compliance (2 tables)

Table	Purpose	POPIA Requirement
audit_log	Change tracking	Section 17 —all data modifications logged
data_processing_register	GDPR entries	Section 17 —lawful basis, retention, safeguards

Foreign Key Relationships

clients	└─> projects (1:N)
	└─> invoices (1:N)
	└─> payments (1:N)
	└─> subscriptions (1:N)
	└─> consultations (converted_to_client_id)
	└─> project_comments (1:N)
admins	└─> consultations (assigned_admin_id)
	└─> messages (replied_by)
	└─> project_comments (1:N)
	└─> audit_log (user_id)
projects	└─> project_milestones (1:N)
	└─> project_files (1:N)

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└─> project_comments (1:N)
└─> invoices (1:N)
└─> consultations (converted_to_project_id)
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invoices → payments (1:N)
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consultations → calculator_leads (1:1, optional)
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Generated Columns

- `invoices.total_amount = amount + tax_amount`
- `invoice_items.total = quantity * unit_price`

Indexes Strategy

- **Search:** email, slug, status on all major tables
- **Date ranges:** created_at, due_date, booking_date, published_at
- **Analytics:** ip_hash, session_id, page_path
- **Foreign keys:** All FK columns indexed for JOIN performance

POPIA Compliance Built-In

- ip_hash instead of raw IP in analytics
- data_processing_register table for Section 17
- audit_log for all data changes
- Consent tracking in consultations and settings
- 7-year retention for financial records
- Right to deletion: cascading deletes on clients → projects, comments, files